Tanish Singh Rajpal

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Reliable quantum information systems: learning, verification, and simulation

EDUCATION

Carnegie Mellon University — *Pittsburgh, PA*
M.S. in Information Technology - Information Security — *Expected May 2027*
Information Networking Institute

NMIMS University — *Mumbai, India*
B.Tech. in Computer Engineering — *May 2025*
GPA: 3.83/4.00 | Valedictorian, graduating class
Relevant coursework: Linear Algebra and Differential Equations, Complex Variables and Transforms, Probability and Statistics, Discrete Mathematics, Quantum and Statistical Physics, Design and Analysis of Algorithms, Theoretical Computer Science, Operations Research, Machine Learning (all A or A+).

RESEARCH EXPERIENCE

Certifying delivered entanglement in quantum networks — *May 2026 - present*
Independent Study, CMU Information Networking Institute | Advisor: Prof. Michael Mattarock
- Established a test-only limit on per-pair certification and developed a two-layer method: an anytime-valid model-based guard, a model-free average-fidelity certificate, and a statistical model check with a post-alarm guarantee.
- Ran pre-registered simulations on grids up to 900 nodes: no below-threshold pairs among 4.2 million simulated deliveries after an alarm. Across the project, 83 of 95 primary hypotheses passed; all 12 failures are reported.
- Re-analysed data from five experiments (Innsbruck, ICFO, USTC, Delft, Oxford) and ran two test-or-keep experiments on one IBM quantum processor. These are component checks, not a physical-network demonstration. Manuscript in preparation; code and evidence.

Differentiable quantum-chemical models (DFTBML3) — *2026 - present*
Independent Study, CMU Chemistry | Advisor: Prof. David Yaron
- Working on differentiable density-functional tight-binding models for chemical Hamiltonians, preserving physical structure while testing trainable components against established implementations. Results in progress.

A hybrid architecture for post-quantum EAP-TLS — *Aug 2026 - present*
Master's Thesis, CMU | Advisors: Prof. Pedro Bustamante and Prof. Michael Mattarock; Reader: Prof. Patrick Tague
- Investigating whether oversized ML-DSA certificates can cause Path MTU Discovery blackholes in WPA-Enterprise 802.1X authentication.
- Designing an ns-3 evaluation of an edge QUIC proxy and QKD-secured wired backbone, with hypotheses covering handshake latency, RADIUS compute, IPv4/IPv6 behavior, and key-delivery capacity. Evaluation in progress.

SELECTED QUANTUM PROJECTS

Phase Hunter - mapping quantum spin phases under noise — *Sep 2026*
Q-SITE Hacks 2026, Quantum Coalition Scientific Track runner-up | Team Popeyes
- With Paul Jeon and Sumedhaa Ruhil, built a playable phase-boundary search backed by reproducible PennyLane simulations. Compared clean, noisy, and error-mitigated phase maps under limited measurement budgets. Code and notebook.

Quantum portfolio optimization using QAOA — *Apr 2026*
Team Lead, Yale YQuantum Hackathon | Third place
- Led the Markowitz-to-QUBO-to-Ising formulation and QAOA circuit design for an insurance portfolio optimization prototype. Repository.

PEER-REVIEWED PUBLICATIONS

Journal article

Rajpal, T. S., and Naithani, A. (2026). "NeuroCrypt: A Neuro Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission." *IEEE Transactions on Artificial Intelligence*, 7(1), 512-521. doi:10.1109/TAI.2025.3577605

Conference papers

Naithani, A., and Rajpal, T. S. (2025). "Blockchain-Based Digital Health Passport: A Decentralized Framework for Secure and Interoperable Health Data Management." *IEEE ZINC 2025*. doi:10.1109/ZINC65316.2025.11100281

Naithani, A., Rajpal, T. S., and Mistry, V. (2024). "TAV-Cart: Transaction Assistance and Validation Cart System for Effortless Shopping and Checkout." *IEEE ICACCM 2024*. doi:10.1109/ICACCM61117.2024.11059075

Naithani, A., Jain, V., Rajpal, T. S., and Mistry, V. (2024). "AI Based Goalkeeper for Penalty Shot Prediction Using RNN and Pattern Matching Algorithm." *IEEE ISDFS 2024*. doi:10.1109/ISDFS60797.2024.10527307

HONORS AND TRAINING

- Q-SITE Hacks 2026: runner-up, Quantum Coalition Open Challenge (Scientific Track); third place, IBM Quantum Coding Challenge; third place, Overall Coding Challenge; fourth place, IonQ Championship.
- IBM Qiskit Global Summer School 2026 - Quantum Excellence (Advanced). Credential.
- Third place, Yale YQuantum Hackathon (2026); first place, Crypto Quest Cybersecurity Competition (2024).

RESEARCH METHODS AND TOOLS

Methods: quantum-network simulation, pre-registered experiments, model checking, statistical certification, reinforcement learning, PPO, QAOA, QUBO/Ising formulations.

Platforms: NetSquid, ns-3, PennyLane, Qiskit, Python, Git/GitHub.

Security and networking: EAP-TLS, ML-DSA, 802.1X, PMTUD, QKD, Burp Suite, Metasploit.

INDUSTRY EXPERIENCE

Vulnerability Assessment and Penetration Testing Intern *Oct 2023 - Jan 2024*

Newton's Apple, Pune, India

- Identified and supported remediation of three critical vulnerabilities in a confidential Android application; performed static Java code analysis with the development team.

ACADEMIC SERVICE

Student Ambassador, CMU Information Networking Institute *Oct 2025 - present*

Support prospective students and help plan community events.